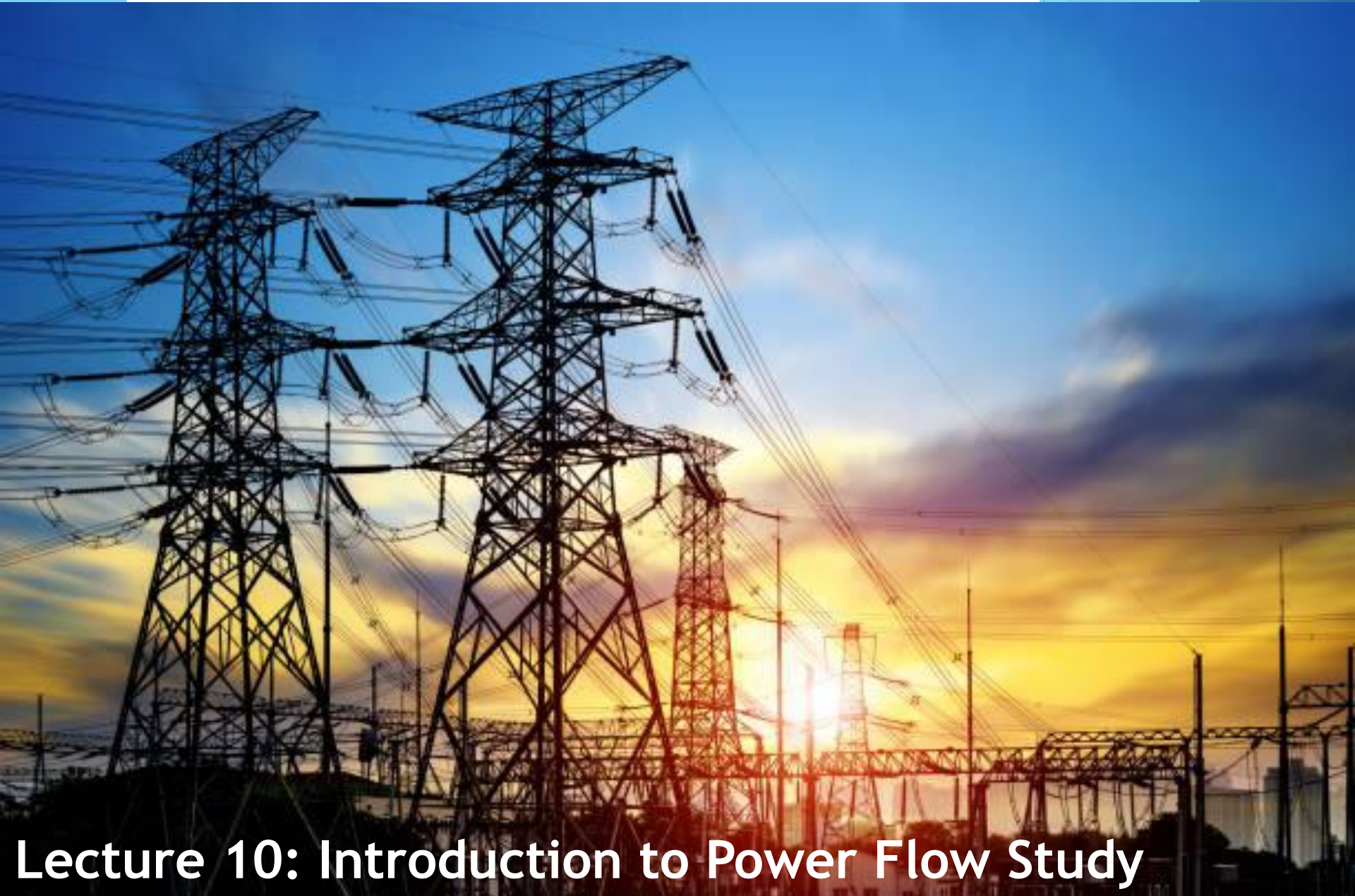




Lecture 10: Introduction to Power Flow Study

EEE 321: Power System I

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Presentation Outline

- ❑ Introduction to Power Flow Study
- ❑ Steps in Power Flow Study
- ❑ Bus Admittance Matrix Y_{bus}
- ❑ Power Flow Equations
- ❑ Comparison between Gauss-Seidel and Newton-Raphson Method
- ❑ Gauss-Seidel Power Flow Solution
- ❑ Examples and Problems

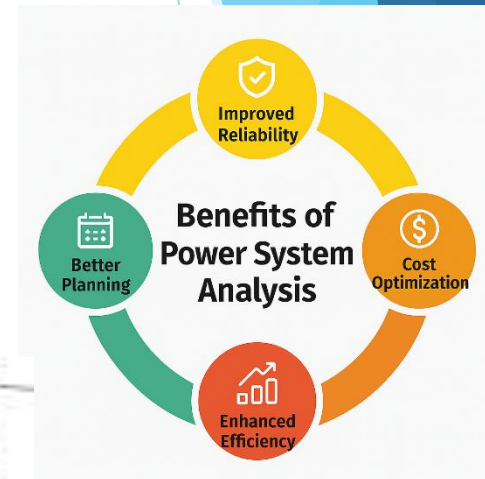
Introduction to Power Flow Study

The primary types of power system analysis include:

- ☒ **Power Flow Analysis (steady-state system performance)**
- ☐ Fault Analysis (examining effects of short circuits)
- ☐ Stability Analysis (system response to disturbances)
- ☐ Power Quality and Harmonic Analysis (evaluating non-linear load distortion).

Other key types include

- ☐ Contingency Analysis (system performance with component failures)
- ☐ Reliability Analysis (system dependability)
- ☐ Optimization Analysis (cost-effective operation)
- ☐ Protection Coordination (setting protective devices).



Introduction to Power Flow Study

- ❑ We seek to analyze the power system performance under steady state conditions.
- ❑ Power flow studies (load flow) is an important component of power system analysis.
- ❑ Necessary for planning, economic scheduling and control of an existing system as well as planning its future expansion.
- ❑ The problem consists of **determining the magnitudes and phase angle of voltages at each bus and active and reactive power flow in each line.**
- ❑ In solving power flow problem, the system is assumed to be operating under balanced conditions and a single-phase model is used.
- ❑ Four quantities are associated with each bus
 - i) voltage magnitude, $|V|$
 - ii) phase angle, δ
 - iii) real power, P
 - iv) reactive power, Q

Introduction to Power Flow Study

The system buses are generally classified into three types:-

1. Load bus (PQ bus) –

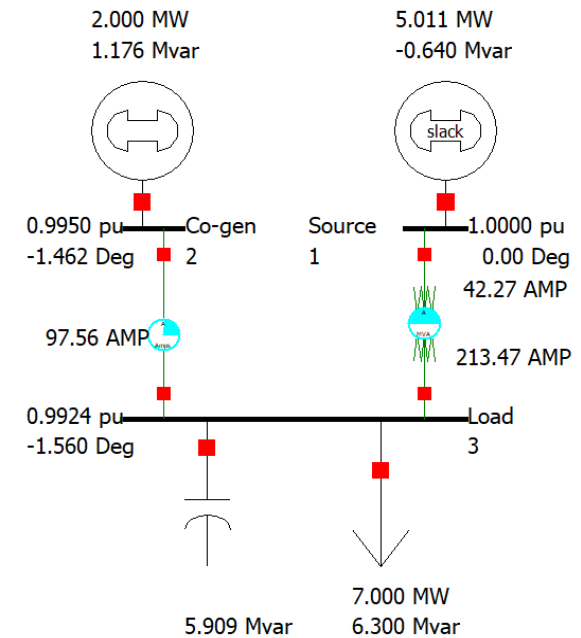
- Buses not having a generator
 - Real and reactive power (**P** and **Q**) are specified
 - Bus voltage magnitude and phase angle (**V** and **δ**) will be calculated.
 - Real and reactive powers supplied to a power system are defined to be positive.
 - Powers consumed from the system are defined to be negative.

2. Generator bus (PV bus) –

- Voltage and real power supplied are specified
- Bus phase angle (**δ**) will be calculated during iteration
- Reactive power will be calculated after the case's solution is found.

3. Slack bus (swing bus) –

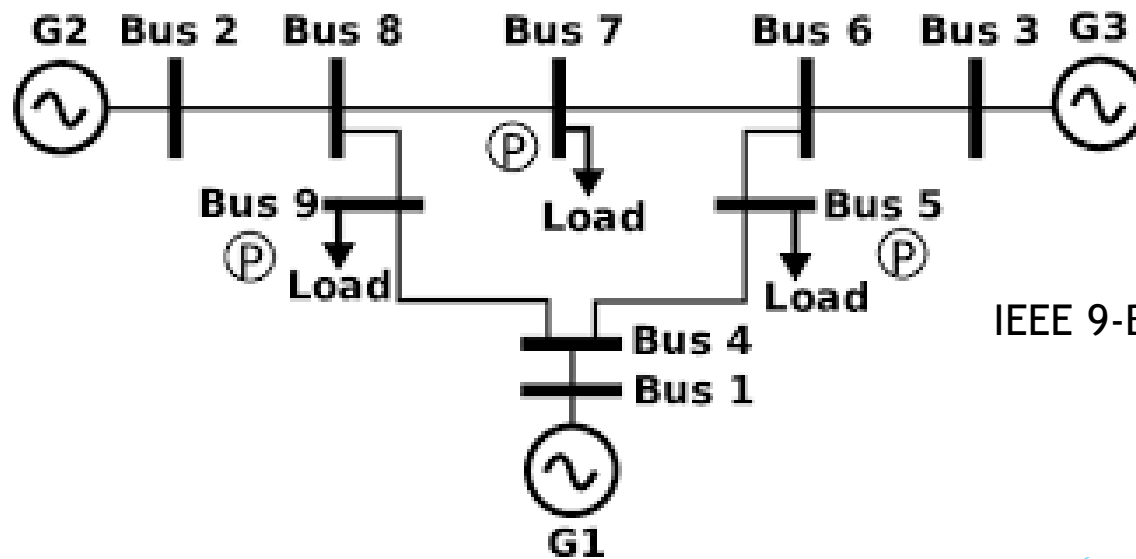
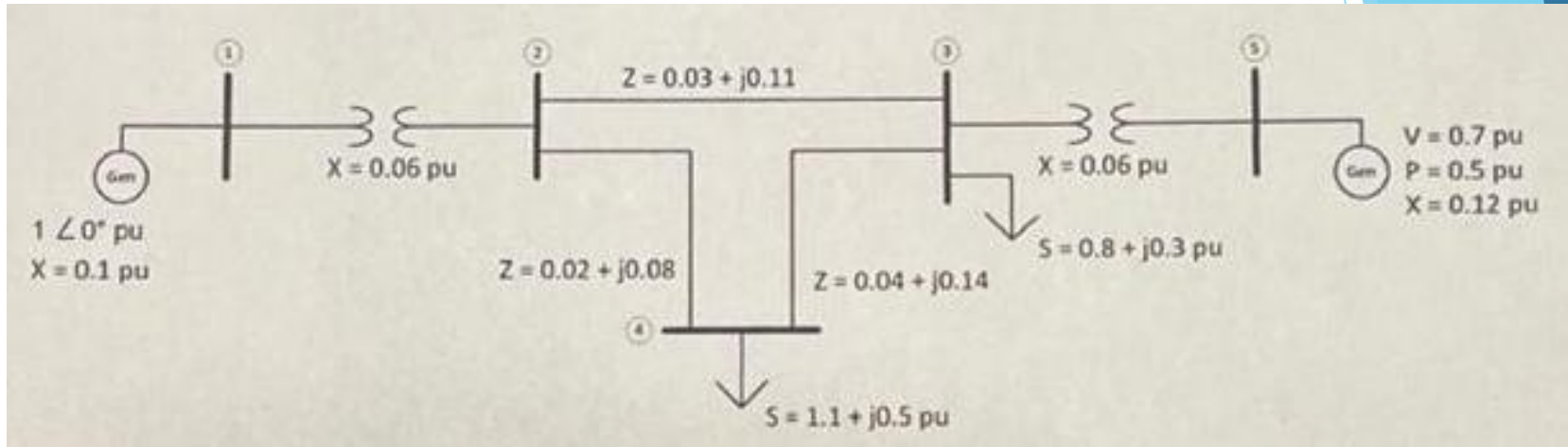
- Special generator bus serving as the reference bus for the power system.
- Voltage is fixed – both magnitude and phase (for instance, $1 \angle 0^\circ$ pu).
- Real and reactive powers are uncontrolled – supplies whatever real or reactive power is necessary to make the power flows in the system balance.



Bus type	Quantities Specified	Quantities to be obtained
Load bus	P, Q	$ V , \delta$
Generator bus	$P, V $	Q, δ
Slack bus	$ V , \delta$	P, Q

Introduction to Power Flow Study

Identify the bus type

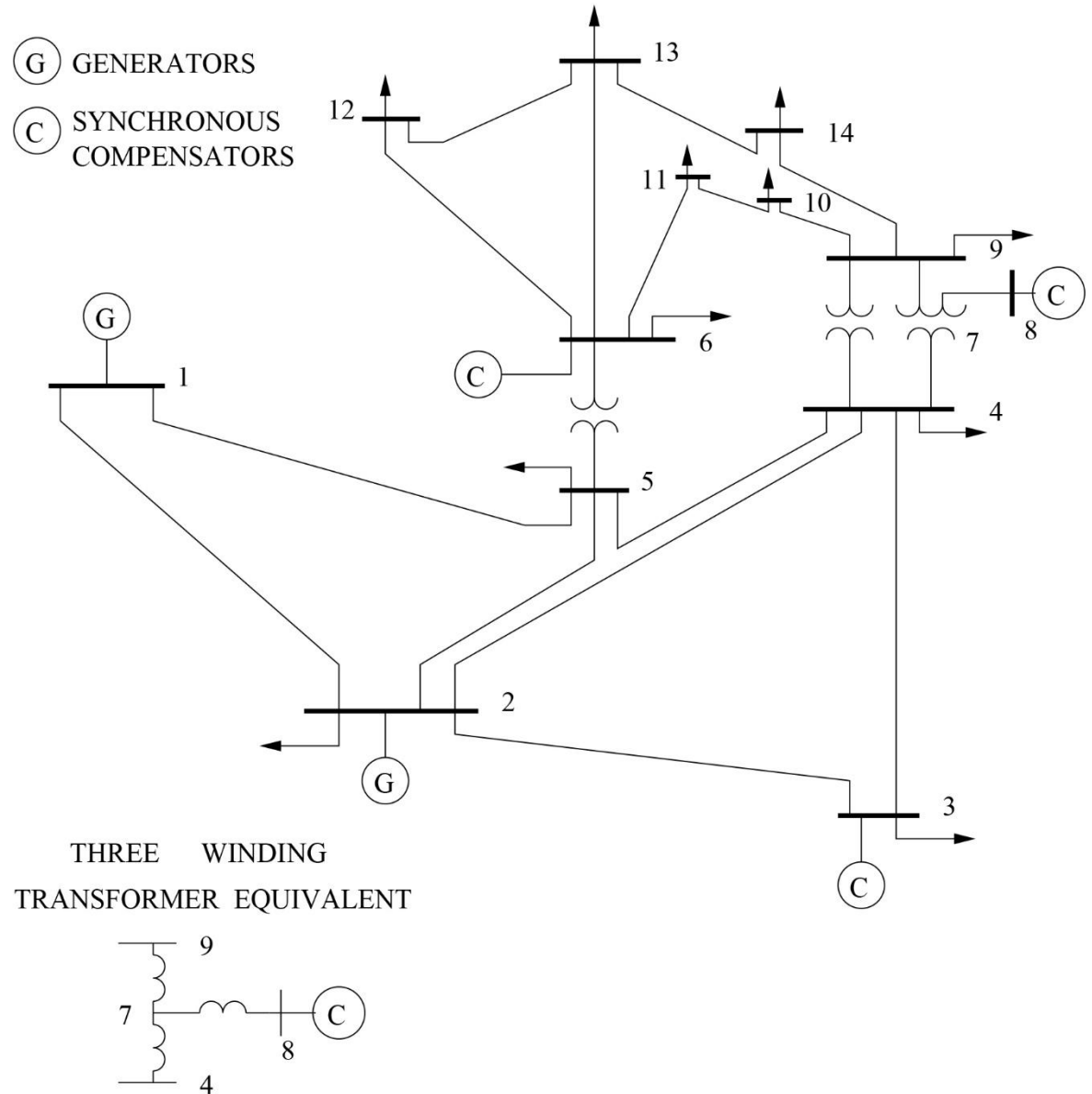


IEEE 9-Bus system

Introduction to Power Flow Study

Identify the bus type

IEEE 14-Bus system

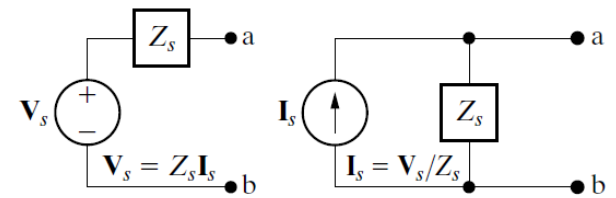


Steps in Power-flow Studies

To find the power-flow solution via iteration:

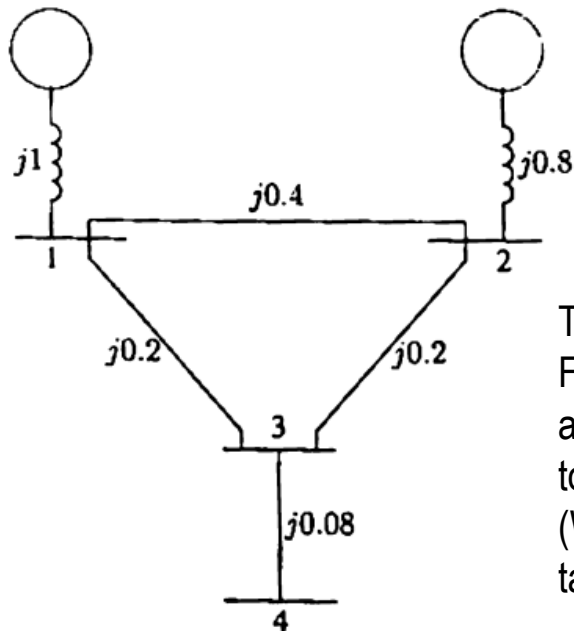
1. Create a bus admittance matrix $\mathbf{Y}_{\text{bus}}$ for the power system;
2. Formulate **Power Flow Equations from Node Voltage Equations**.
3. Make an **initial estimate** for the voltages at each bus in the system;
4. **Iterate to find conditions** that **satisfy** the system's load flow equations.
 - Update the voltage estimate for each bus (one at a time), based on the estimates for the voltages and power flows at every other bus and the values of the bus admittance matrix.
 - Since the voltage at a given bus depends on the voltages at all of the other buses in the system (which are just estimates), the updated voltage will not be correct. However, it will usually be closer to the answer than the original guess.
5. Repeat this process to make the voltages at each bus approaching the correct answers to **within a set tolerance** level.

Bus Admittance Matrix Y_{bus}



In order to obtain the node-voltage equations, consider the simple power system shown in Figure 6.1 where impedances are expressed in per unit on a common MVA base and for simplicity resistances are neglected. Since the nodal solution is based upon Kirchhoff's current law, impedances are converted to admittance, i.e.,

$$y_{ij} = \frac{1}{z_{ij}} = \frac{1}{r_{ij} + jx_{ij}}$$



The circuit has been redrawn in Figure 6.2 in terms of admittances and transformation to current sources. Node 0 (Which is normally ground) is taken as reference.

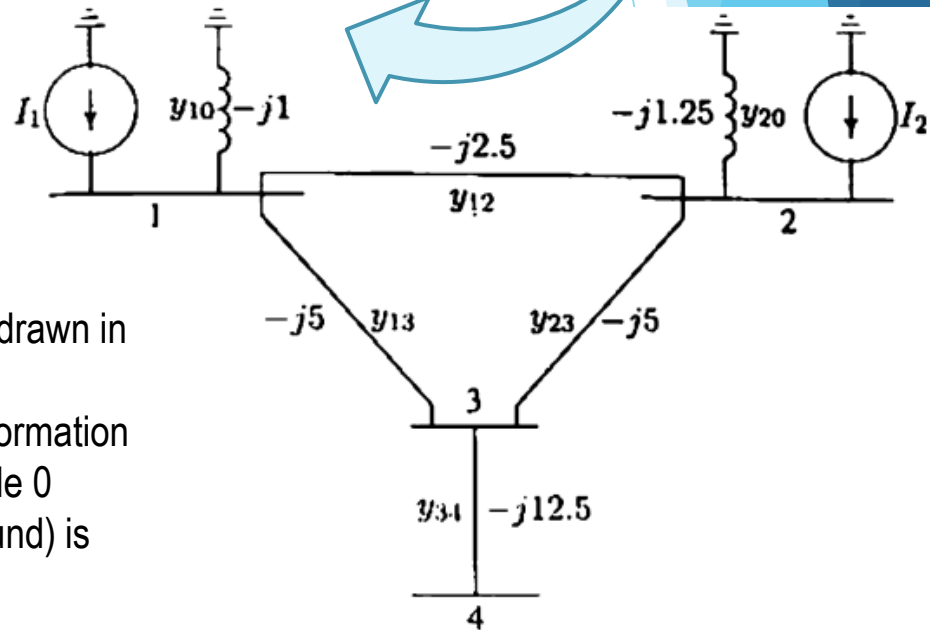
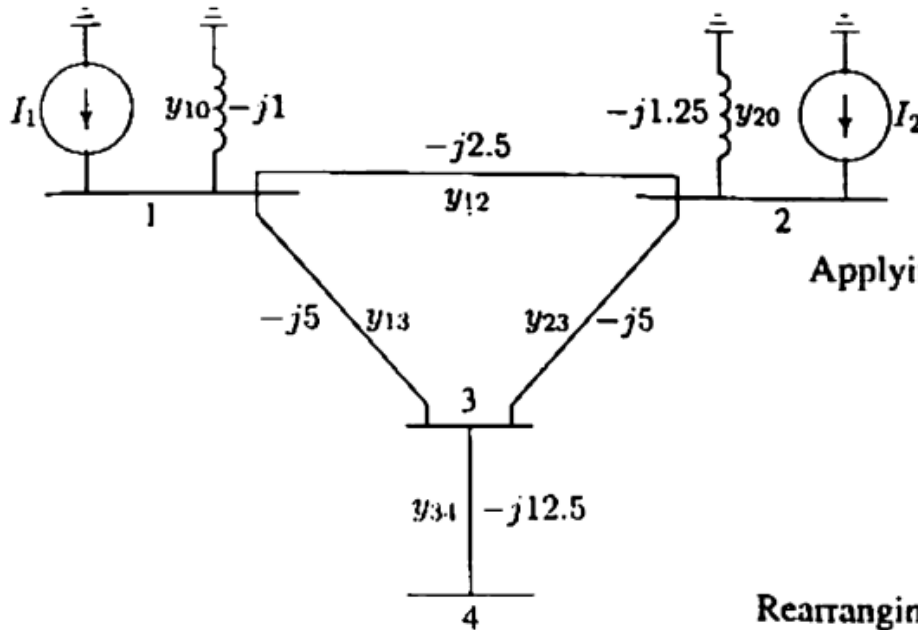


FIGURE 6.1
The impedance diagram of a simple system.

FIGURE 6.2
The admittance diagram for system of Figure 6.1.

Bus Admittance Matrix Y_{bus}



Applying KCL to the independent nodes 1 through 4 results in

$$\begin{aligned} I_1 &= y_{10}V_1 + y_{12}(V_1 - V_2) + y_{13}(V_1 - V_3) \\ I_2 &= y_{20}V_2 + y_{12}(V_2 - V_1) + y_{23}(V_2 - V_3) \\ 0 &= y_{23}(V_3 - V_2) + y_{13}(V_3 - V_1) + y_{34}(V_3 - V_4) \\ 0 &= y_{34}(V_4 - V_3) \end{aligned}$$

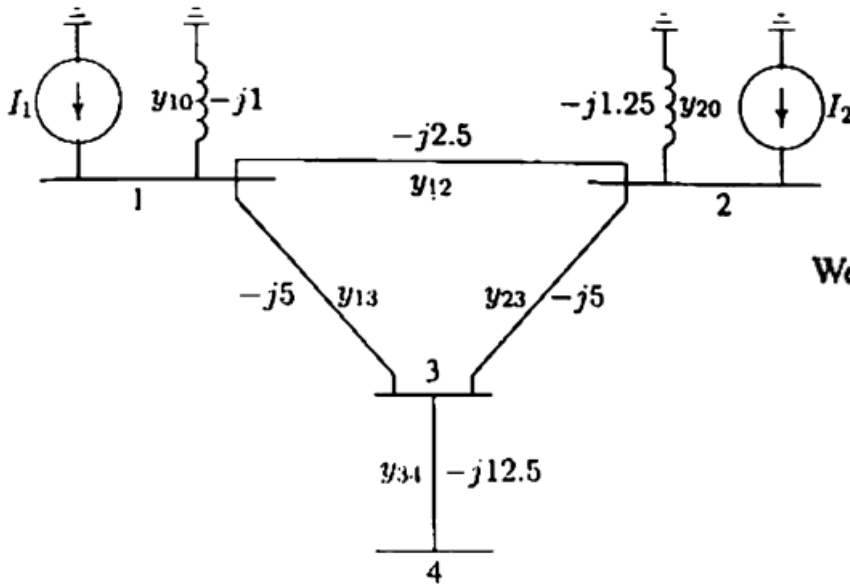
Rearranging these equations yields

$$\begin{aligned} I_1 &= (y_{10} + y_{12} + y_{13})V_1 - y_{12}V_2 - y_{13}V_3 \\ I_2 &= -y_{12}V_1 + (y_{20} + y_{12} + y_{23})V_2 - y_{23}V_3 \\ 0 &= -y_{13}V_1 - y_{23}V_2 + (y_{13} + y_{23} + y_{34})V_3 - y_{34}V_4 \\ 0 &= -y_{34}V_3 + y_{34}V_4 \end{aligned}$$

FIGURE 6.2

The admittance diagram for system of Figure 6.1.

Bus Admittance Matrix Y_{bus}



We introduce the following admittances

$$Y_{11} = y_{10} + y_{12} + y_{13}$$

$$Y_{22} = y_{20} + y_{12} + y_{23}$$

$$Y_{33} = y_{13} + y_{23} + y_{34}$$

$$Y_{44} = y_{34}$$

$$Y_{12} = Y_{21} = -y_{12}$$

$$Y_{13} = Y_{31} = -y_{13}$$

$$Y_{23} = Y_{32} = -y_{23}$$

$$Y_{34} = Y_{43} = -y_{34}$$

FIGURE 6.2
The admittance diagram for system of Figure 6.1.

The node equation reduces to

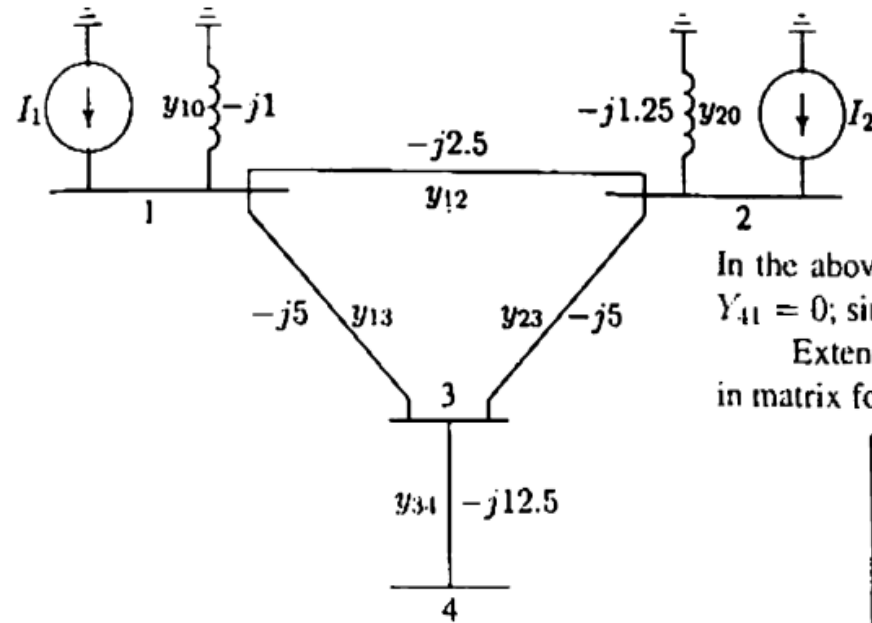
$$I_1 = Y_{11}V_1 + Y_{12}V_2 + Y_{13}V_3 + Y_{14}V_4$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3 + Y_{24}V_4$$

$$I_3 = Y_{31}V_1 + Y_{32}V_2 + Y_{33}V_3 + Y_{34}V_4$$

$$I_4 = Y_{41}V_1 + Y_{42}V_2 + Y_{43}V_3 + Y_{44}V_4$$

Bus Admittance Matrix Y_{bus}



$$\begin{aligned} I_1 &= Y_{11}V_1 + Y_{12}V_2 + Y_{13}V_3 + Y_{14}V_4 \\ I_2 &= Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3 + Y_{24}V_4 \\ I_3 &= Y_{31}V_1 + Y_{32}V_2 + Y_{33}V_3 + Y_{34}V_4 \\ I_4 &= Y_{41}V_1 + Y_{42}V_2 + Y_{43}V_3 + Y_{44}V_4 \end{aligned}$$

In the above network, since there is no connection between bus 1 and 4, $Y_{14} = Y_{41} = 0$; similarly $Y_{24} = Y_{42} = 0$.

Extending the above relation to an n bus system, the node-voltage equation in matrix form is

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_i \\ \vdots \\ I_n \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & \dots & Y_{1i} & \dots & Y_{1n} \\ Y_{21} & Y_{22} & \dots & Y_{2i} & \dots & Y_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ Y_{i1} & Y_{i2} & \dots & Y_{ii} & \dots & Y_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ Y_{n1} & Y_{n2} & \dots & Y_{ni} & \dots & Y_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_i \\ \vdots \\ V_n \end{bmatrix} \quad (6.1)$$

or

$$\mathbf{I}_{bus} = \mathbf{Y}_{bus} \mathbf{V}_{bus} \quad (6.2)$$

where $\mathbf{I}_{bus}$ is the vector of the injected bus currents (i.e., external current sources). The current is positive when flowing towards the bus, and it is negative if flowing away from the bus. $\mathbf{V}_{bus}$ is the vector of bus voltages measured from the reference node (i.e., node voltages). $\mathbf{Y}_{bus}$ is known as the *bus admittance matrix*. The diagonal element of each node is the sum of admittances connected to it. It is known as the *self-admittance* or *driving point admittance*, i.e.,

$$Y_{ii} = \sum_{j=0}^n y_{ij} \quad j \neq i \quad (6.3)$$

Bus Admittance Matrix Y_{bus}

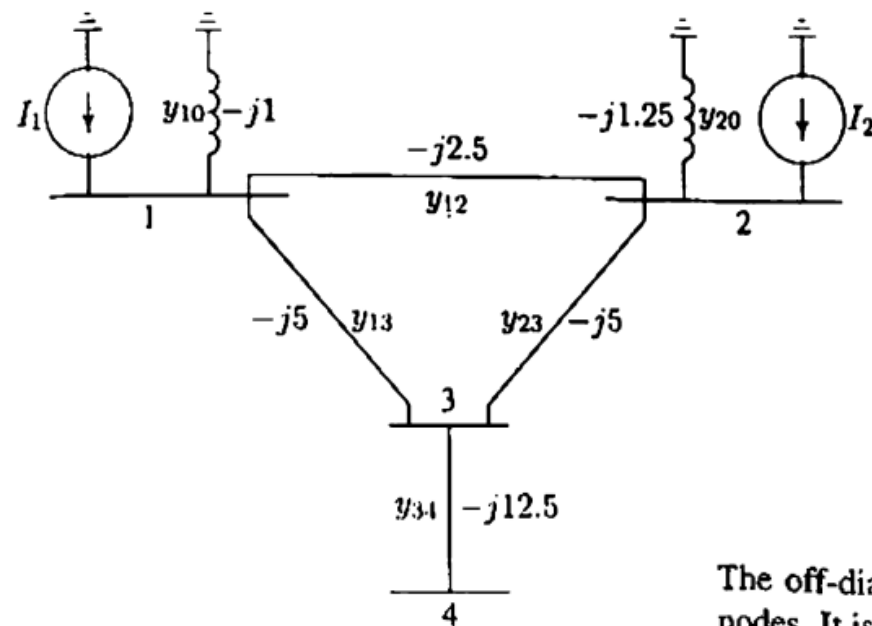


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or

$$I_{bus} = Y_{bus} V_{bus} \quad (6.2)$$

$$Y_{ii} = \sum_{j=1}^n y_{ij} \quad j \neq i \quad (6.3)$$

The off-diagonal element is equal to the negative of the admittance between the nodes. It is known as the *mutual admittance* or *transfer admittance*, i.e.,

$$Y_{ij} = Y_{ji} = -y_{ij} \quad (6.4)$$

The inverse of the bus admittance matrix is known as the *bus impedance matrix* Z_{bus} .

Based on (6.3) and (6.4), the bus admittance matrix for the network in Figure 6.2 obtained by inspection is

$$Y_{bus} = \begin{bmatrix} -j8.50 & j2.50 & j5.00 & 0 \\ j2.50 & -j8.75 & j5.00 & 0 \\ j5.00 & j5.00 & -j22.50 & j12.50 \\ 0 & 0 & j12.50 & -j12.50 \end{bmatrix}$$

Bus Admittance Matrix Y_{bus}

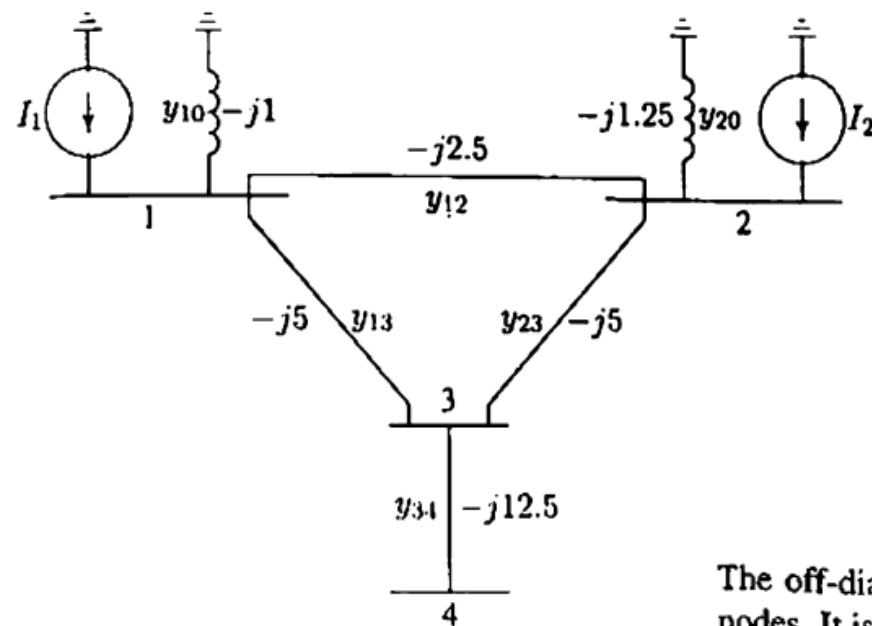


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or

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Power Flow Equations

The basic equation for power-flow analysis is derived from the nodal analysis equations for the power system:

Consider a typical bus of a power system network as shown in Figure 6.7. Transmission lines are represented by their equivalent π models where impedances have been converted to per unit admittances on a common MVA base.

Application of KCL to this bus results in

$$\begin{aligned} I_i &= y_{i0}V_i + y_{i1}(V_i - V_1) + y_{i2}(V_i - V_2) + \cdots + y_{in}(V_i - V_n) \\ &= (y_{i0} + y_{i1} + y_{i2} + \cdots + y_{in})V_i - y_{i1}V_1 - y_{i2}V_2 - \cdots - y_{in}V_n \quad (6.23) \end{aligned}$$

$$= V_i \sum_{j=0} y_{ij} - \sum_{j=1} y_{ij}V_j \quad j \neq i \quad (6.24)$$

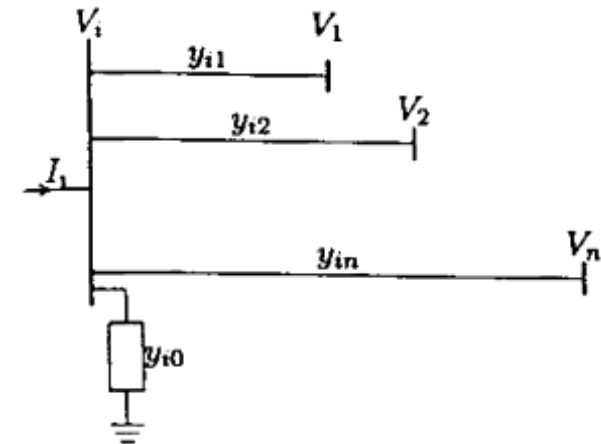


FIGURE 6.7
A typical bus of the power system.

The real and reactive power at bus i is

$$P_i + jQ_i = V_i I_i^* \quad (6.25)$$

or

$$I_i = \frac{P_i - jQ_i}{V_i^*} \quad (6.26)$$

Substituting for I_i in (6.24) yields

$$\frac{P_i - jQ_i}{V_i^*} = V_i \sum_{j=0} y_{ij} - \sum_{j=1} y_{ij}V_j \quad j \neq i \quad (6.27)$$

From the above relation, the mathematical formulation of the power flow problem results in a system of algebraic nonlinear equations which must be solved by iterative techniques.

From this relationship, the mathematical formulation of the power flow problem results in a system of algebraic nonlinear equations which must be solved by iterative techniques.

1. Gauss-Seidel Method
2. Newton-Raphson Method

Comparison between Gauss-Seidel and Newton-Raphson Method

Comparison between Gauss-Seidel And Newton-Raphson method

Feature	Gauss-Seidel (G-S) Method	Newton-Raphson (N-R) Method
Convergence	Slow, linear convergence; requires more iterations	Faster, quadratic convergence; requires fewer iterations
Accuracy	Less accurate compared to N-R	More accurate, providing more reliable results
System Suitability	Best for small power systems	Suitable for large power systems
Complexity	Simpler calculations and programming tasks	More complex programming and algorithm
Memory Requirement	Less memory required	More complex memory requirements
Computational Time	Lower computational time per iteration but higher total time due to more iterations	Higher computational time per iteration but lower total time due to fewer iterations
Key Advantage	Ease of implementation, simple logic	Faster convergence, reliability, and high accuracy for large systems
Key Disadvantage	Slow convergence for larger systems, can have issues with long radial lines	Higher computational effort and complexity per iteration

Gauss-Seidel Power Flow Solution

- In power flow study, it is essential to solve the set of non-linear equations for **two unknown variables at each node**.
- In Gauss-Seidel method, V_i is solved and the iterative sequences become

$$V_i^{(k+1)} = \frac{\frac{P_i^{sch} - jQ_i^{sch}}{V_i^{*(k)}} + \sum y_{ij} V_j^{(k)}}{\sum y_{ij}} \quad j \neq i$$

where,

y_{ij} shown in lowercase letters is the **actual admittance** in per unit
 P_i^{sch} and Q_i^{sch} are the **net real** and **reactive powers** in per unit

- In KCL, current entering **bus i** was **assumed positive**. Thus, for buses where **real and reactive powers are injected into the bus**, such as **generator buses**, P_i^{sch} and Q_i^{sch} have **positive** values.
- For **load buses** where real and reactive powers are flowing **away** from the bus, P_i^{sch} and Q_i^{sch} have **negative** values.

Gauss-Seidel Power Flow Solution

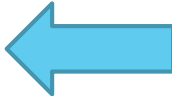
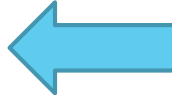
- Rewriting the power flow equation in terms of the **bus admittance matrix (Y-bus)**.
- Since the **off-diagonal elements** of the bus admittance matrix Y_{bus} shown by uppercase letters, $Y_{ij} = -y_{ij}$ and the **diagonal elements** are $Y_{ii} = \sum y_{ij}$, the equation becomes

$$V_i^{[k+1]} = \frac{\frac{P_i^{[sch]} - jQ_i^{[sch]}}{V_i^{*[k]}} - \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]}}{Y_{ii}}$$

$$P_i^{[k+1]} = \Re \left\{ V_i^{*[k]} \left[V_i^{[k]} Y_{ii} + \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]} \right] \right\} \quad Q_i^{[k+1]} = -\Im \left\{ V_i^{*[k]} \left[V_i^{[k]} Y_{ii} + \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]} \right] \right\}$$

- Y_{ii} includes the admittance to ground of line charging susceptance and any other fixed admittance to ground.
- Since both components (**V and δ**) are specified for slack bus, there are **2 (n-1) equations** which must be solved iteratively.
- Under normal operating conditions, **|V| of buses** are in neighborhood of 1.0 per unit or close to the **|V| of the slack buses**.
- **|V| of load buses** are **lower** than the **slack bus** value, depending on reactive power demand whereas the **scheduled voltage at the generator buses** are higher.

Gauss-Seidel Power Flow Solution

- The δ of the load buses are below the reference angle in accordance to power demand whereas δ of the generator buses may be above the reference value depending on the amount of real power flowing into the bus.
- Thus, for Gauss-Seidel method, an initial voltage estimate of $1.0+j0.0$ for unknown voltage is satisfactory and the converged solution correlates with actual operating states. 
- For P-Q buses (load buses), P_i^{sch} and Q_i^{sch} are known. Starting with an initial estimate, $V_i^{(k+1)}$ is solved for the real and imaginary components of voltage. 
- For P-V buses (regulated buses), P_i^{sch} and $|V_i|$ are specified, $Q_i^{(k+1)}$ is solved, then used in $V_i^{(k+1)}$. 
- Since $|V_i|$ is specified, only imaginary part of $V_i^{(k+1)}$ is retained. And its real part is selected in order to satisfy

$$e_i^{[k+1]} = \sqrt{|V_i^{[sch]}|^2 - (f_i^{[k+1]})^2}$$

where $e_i^{(k+1)}$ and $f_i^{(k+1)}$ are the real and imaginary components of the voltage $V_i^{(k+1)}$ in the iterative sequence.

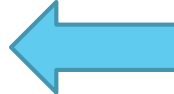
Gauss-Seidel Power Flow Solution

- The rate of convergence is increased by applying **acceleration factor (α)** to the approximate solution obtained from each iteration.

$$V_i^{(k+1)} = V_i^{(k)} + \alpha (V_{i\text{ cal}}^{(k)} - V_i^{(k)})$$

- α is the **acceleration factor** and is in the range of **1.3 to 1.7**.
- The process is continued until changes in the real and imaginary components of bus voltages between successive iterations are within **specified accuracy**.

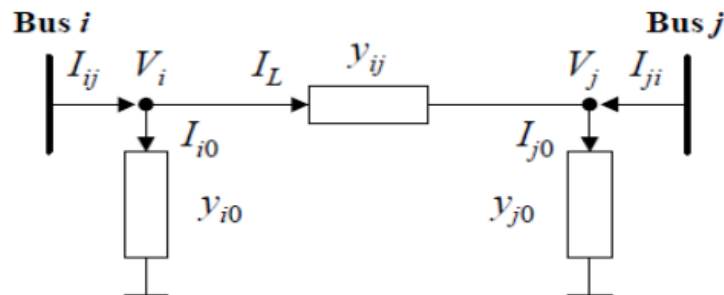
$$\begin{aligned} |e_i^{(k+1)} - e_i^{(k)}| &\leq \varepsilon \\ |f_i^{(k+1)} - f_i^{(k)}| &\leq \varepsilon \end{aligned}$$

- A voltage in accuracy is in the range of **0.00001 to 0.00005 per unit** is satisfactory. 
- In practice, the method for determining the completion of a solution is based on an accuracy index set up on the power mismatch.
- The iteration continues until the magnitude of the largest element in the ΔP and ΔQ columns is less than specified value.
- Normally, power mismatch accuracy is **0.001 per unit**.

Gauss-Seidel Power Flow Solution

Line Flows and Losses

- After solving for bus voltages and angles, **power flows and losses** in the network branches are calculated.
- Consider the line connecting the **two buses, bus i and bus j** in transmission line model below. The line current, I_{ij} measured at **bus i** and defined positive in the direction.



• current and power flows:

$i \rightarrow j$

$$I_{ij} = I_L + I_{i0} = y_{ij}(V_i - V_j) + y_{i0} V_i$$

$$S_{ij} = V_i I_{ij}^* = V_i^2 (y_{ij} + y_{i0})^* - V_i y_{ij}^* V_j^*$$

$j \rightarrow i$

$$I_{ji} = -I_L + I_{j0} = y_{ij}(V_j - V_i) + y_{j0} V_j$$

$$S_{ji} = V_j I_{ji}^* = V_j^2 (y_{ij} + y_{j0})^* - V_j y_{ij}^* V_i^*$$

• power loss:

$$S_{Loss\ ij} = S_{ij} + S_{ji}$$

Gauss-Seidel Power Flow Solution

Example#1

A simple power system has 4 buses, 5 transmission lines, 1 generator, and 3 loads. Series per-unit impedances are given. Determine the Bus Admittance Matrix Y_{bus} .

Solution:

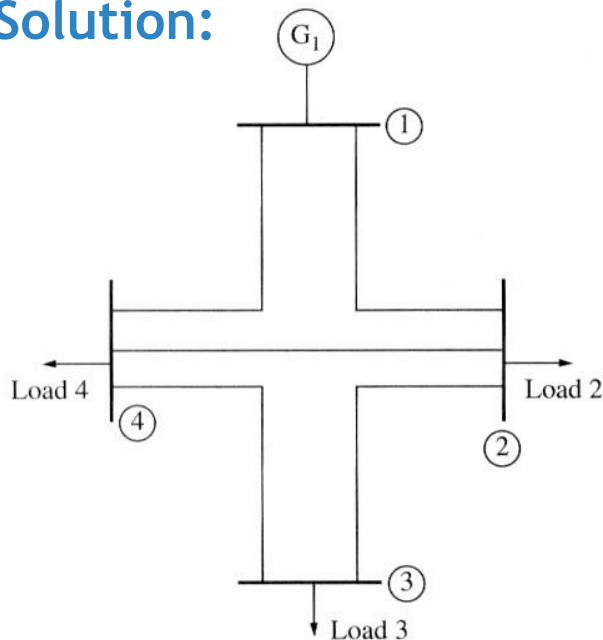


Table of Busses:

Bus 1	Slack bus
Bus 2	Load bus
Bus 3	Load bus
Bus 4	Load bus

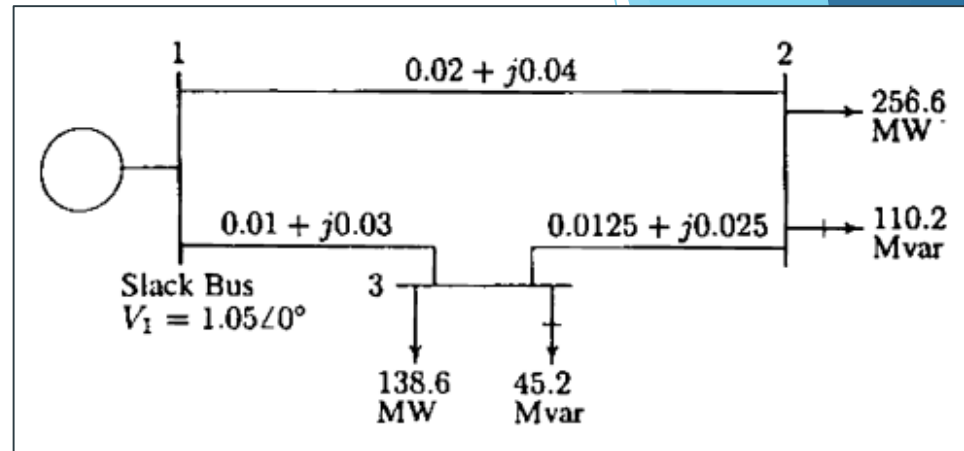
line #	Bus to bus	Series Z (pu)	Series Y (pu)
1	1-2	$0.1+j0.4$	$0.5882-j2.3529$
2	2-3	$0.1+j0.5$	$0.3846-j1.9231$
3	2-4	$0.1+j0.4$	$0.5882-j2.3529$
4	3-4	$0.5+j0.2$	$1.1765-j4.7059$
5	4-1	$0.5+j0.2$	$1.1765-j4.7059$

$$Y_{bus} = \begin{bmatrix} 1.7647 - j7.0588 & -0.5882 + j2.3529 & 0 & -1.1765 + j4.7059 \\ -0.5882 + j2.3529 & 1.5611 - j6.6290 & -0.3846 + j1.9231 & -0.5882 + j2.3529 \\ 0 & -0.3846 + j1.9231 & 1.5611 - j6.6290 & -1.1765 + j4.7059 \\ -1.1765 + j4.7059 & -0.5882 + j2.3529 & -1.1765 + j4.7059 & 2.9412 - j11.7647 \end{bmatrix}$$

Gauss-Seidel Power Flow Solution

Example#2

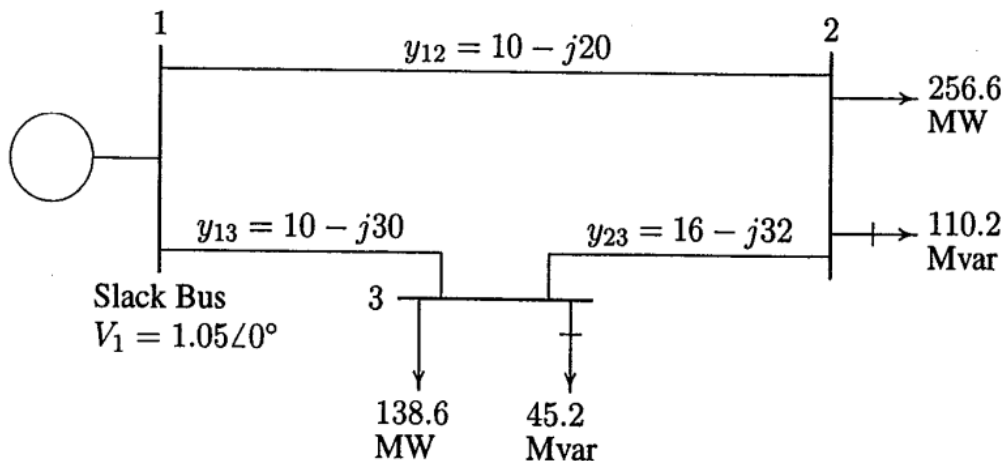
- Using Gauss-Seidel method, find the phasor values of the voltage at load buses 2 and 3 (P-Q buses) to 4 decimal places.
- Find slack bus real and reactive power.
- Determine the line flows and line losses.
- Construct a power flow diagram showing the direction of line flow.



Solution:

- Line impedances are converted to admittances

$$y_{12} = \frac{1}{0.02 + j0.04} = 10 - j20, \quad y_{13} = 10 - j30, \quad y_{23} = 16 - j32$$



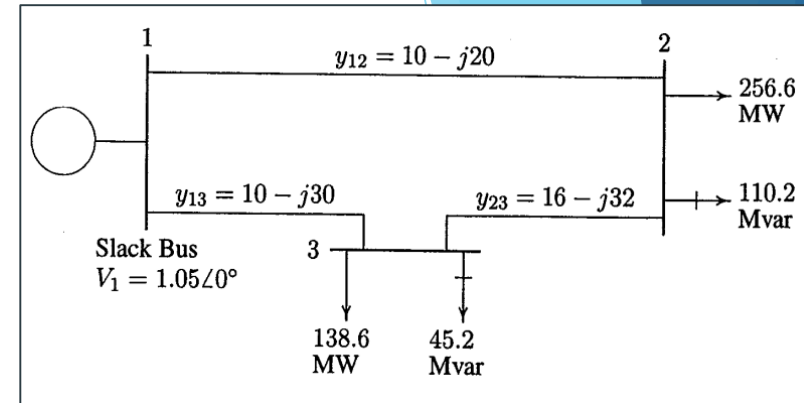
Gauss-Seidel Power Flow Solution

Solution:

At the P-Q buses, the complex loads expressed in per units are

$$S_2^{sch} = -\frac{(256.6 + j110.2)}{100} = -2.566 - j1.102 \text{ pu}$$

$$S_3^{sch} = -\frac{(138.6 + j45.2)}{100} = -1.386 - j0.452 \text{ pu}$$



Bus 1 is taken as reference bus (slack bus). Starting from an initial estimate of

$$V_2^{(0)} = 1.0 + j0.0 \text{ and } V_3^{(0)} = 1.0 + j0.0$$

$$\begin{aligned} V_2^{(1)} &= \frac{\frac{P_2^{sch} - jQ_2^{sch}}{V_2^{*(0)}} + y_{12}V_1 + y_{23}V_3^{(0)}}{y_{12} + y_{23}} \\ &= \frac{\frac{-2.566 + j1.102}{1.0 - j0} + (10 - j20)(1.05 + j0) + (16 - j32)(1.0 + j0)}{(26 - j52)} \\ &= 0.9825 - j0.0310 \text{ pu} \end{aligned}$$

$$\begin{aligned} V_3^{(1)} &= \frac{\frac{P_3^{sch} - jQ_3^{sch}}{V_3^{*(0)}} + y_{13}V_1 + y_{23}V_2^{(1)}}{y_{13} + y_{23}} \\ &= \frac{\frac{-1.386 + j0.452}{1.0 - j0} + (10 - j30)(1.05 + j0) + (16 - j32)(0.9825 - j0.0310)}{(26 - j62)} \\ &= 1.0011 - j0.0353 \end{aligned}$$

Gauss-Seidel Power Flow Solution

Solution:

The process is continued and a solution is converged with an accuracy of 5×10^{-5} per unit in seven iterations as given below.

$$V_2^{(3)} = 0.9808 - j0.0578$$

$$V_3^{(3)} = 1.0004 - j0.0488$$

$$V_2^{(4)} = 0.9803 - j0.0594$$

$$V_3^{(4)} = 1.0002 - j0.0497$$

$$V_2^{(5)} = 0.9801 - j0.0598$$

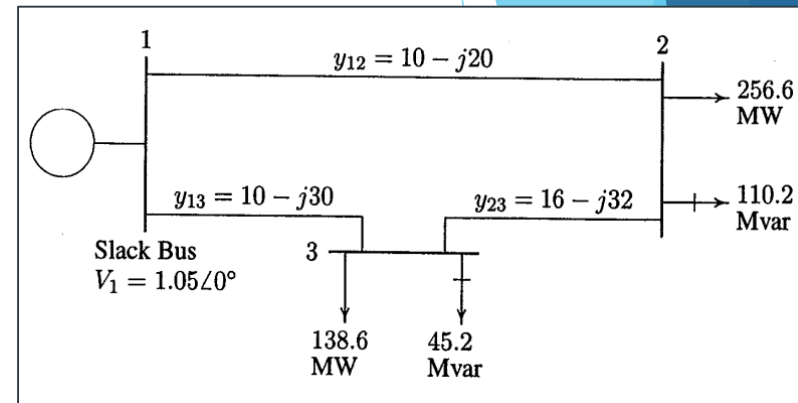
$$V_3^{(5)} = 1.0001 - j0.0499$$

$$V_2^{(6)} = 0.9801 - j0.0599$$

$$V_3^{(6)} = 1.0000 - j0.0500$$

$$V_2^{(7)} = 0.9800 - j0.0600$$

$$V_3^{(7)} = 1.0000 - j0.0500$$



The final solution is

$$V_2 = 0.9800 - j0.0600 = 0.98183 \angle -3.5035^\circ \text{ pu}$$

$$V_3 = 1.0000 - j0.0500 = 1.00125 \angle -2.8624^\circ \text{ pu}$$

Gauss-Seidel Power Flow Solution

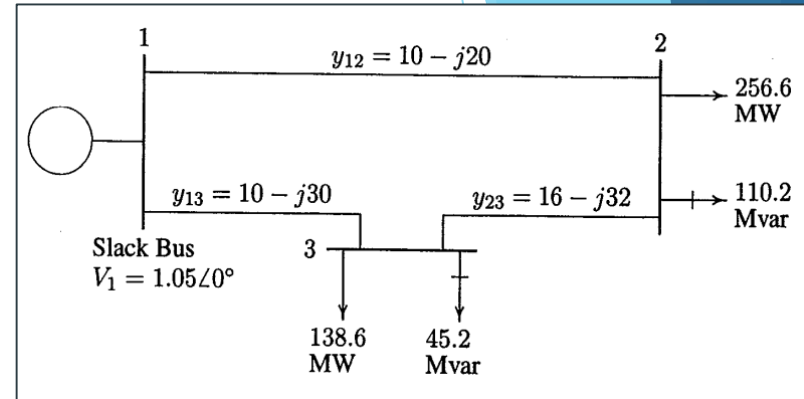
Solution:

b) The slack bus is

$$\begin{aligned} P_1 - jQ_1 &= V_1^* [V_1(y_{12} + y_{13}) - (y_{12}V_2 + y_{13}V_3)] \\ &= 1.05[1.05(20 - j50) - (10 - j20)(0.98 - j0.06) - \\ &\quad (10 - j30)(1.0 - j0.05)] \\ &= 4.095 - j1.890 \end{aligned}$$

or the slack bus real and reactive powers are

$$P_1 = 4.095 \text{ pu} = 409.5 \text{ MW and } Q_1 = 1.890 \text{ pu} = 189.0 \text{ Mvar.}$$



c) To find the line flows, first find the line currents are computed. With the line charging capacitors neglected, the line currents are

$$I_{12} = y_{12}(V_1 - V_2) = (10 - j20)[(1.05 + j0) - (0.98 - j0.06)] = 1.9 - j0.8$$

$$I_{21} = -I_{12} = -1.9 + j0.8$$

$$I_{13} = y_{13}(V_1 - V_3) = (10 - j30)[(1.05 + j0) - (1.0 - j0.05)] = 2.0 - j1.0$$

$$I_{31} = -I_{13} = -2.0 + j1.0$$

$$I_{23} = y_{23}(V_2 - V_3) = (16 - j32)[(0.98 - j0.06) - (1 - j0.05)] = -.64 + j.48$$

$$I_{32} = -I_{23} = 0.64 - j0.48$$

Gauss-Seidel Power Flow Solution

Solution:

The line flows are

$$S_{12} = V_1 I_{12}^* = (1.05 + j0.0)(1.9 + j0.8) = 1.995 + j0.84 \text{ pu} \\ = 199.5 \text{ MW} + j84.0 \text{ Mvar}$$

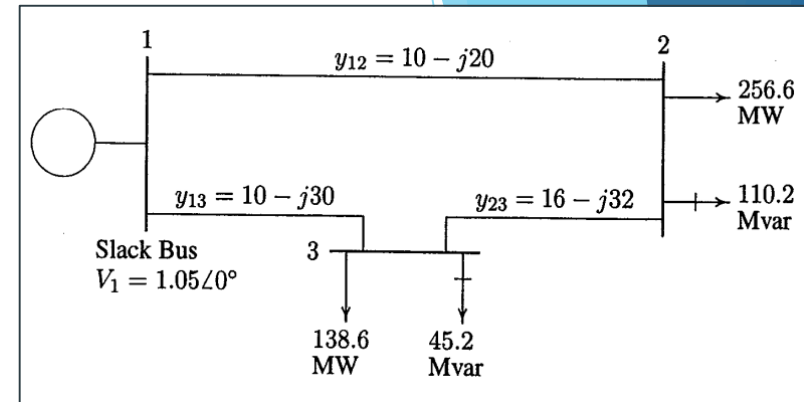
$$S_{21} = V_2 I_{21}^* = (0.98 - j0.06)(-1.9 - j0.8) = -1.91 - j0.67 \text{ pu} \\ = -191.0 \text{ MW} - j67.0 \text{ Mvar}$$

$$S_{13} = V_1 I_{13}^* = (1.05 + j0.0)(2.0 + j1.0) = 2.1 + j1.05 \text{ pu} \\ = 210.0 \text{ MW} + j105.0 \text{ Mvar}$$

$$S_{31} = V_3 I_{31}^* = (1.0 - j0.05)(-2.0 - j1.0) = -2.05 - j0.90 \text{ pu} \\ = -205.0 \text{ MW} - j90.0 \text{ Mvar}$$

$$S_{23} = V_2 I_{23}^* = (0.98 - j0.06)(-0.656 + j0.48) = -0.656 - j0.432 \text{ p} \\ = -65.6 \text{ MW} - j43.2 \text{ Mvar}$$

$$S_{32} = V_3 I_{32}^* = (1.0 - j0.05)(0.64 + j0.48) = 0.664 + j0.448 \text{ pu} \\ = 66.4 \text{ MW} + j44.8 \text{ Mvar}$$



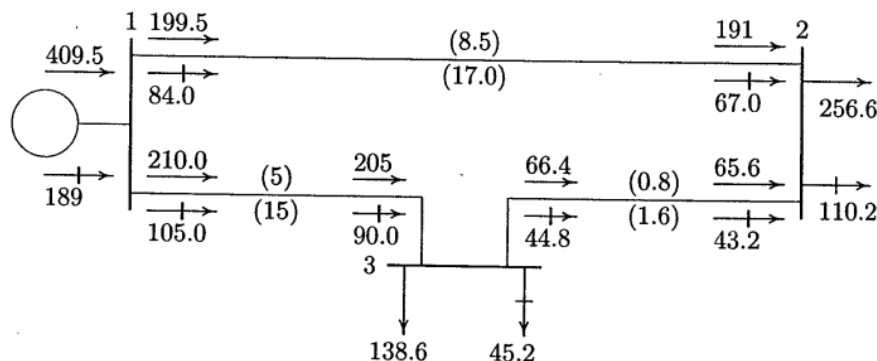
The line losses are

$$S_{L12} = S_{12} + S_{21} = 8.5 \text{ MW} + j17.0 \text{ Mvar}$$

$$S_{L13} = S_{13} + S_{31} = 5.0 \text{ MW} + j15.0 \text{ Mvar}$$

$$S_{L23} = S_{23} + S_{32} = 0.8 \text{ MW} + j1.60 \text{ Mvar}$$

d) The power flow diagram is shown below:-



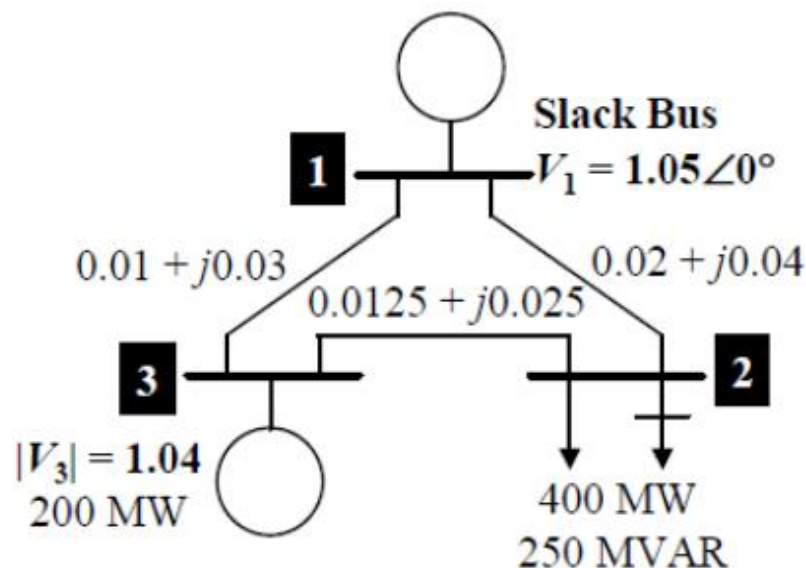
Gauss-Seidel Power Flow Solution

Problem#1

Figure below shows the single-line diagram of three-bus power system with generator at buses 1 and 3. The magnitude of voltage at bus 1 is adjusted to 1.05 per unit. Voltage magnitude at bus 3 is fixed at 1.04 per unit with a real power generation at 200 MW. A load consisting of 400 MW and 250 Mvar is taken from bus 2. Line impedances are marked in per unit on a 100 MVA base and the line charging susceptances are neglected.

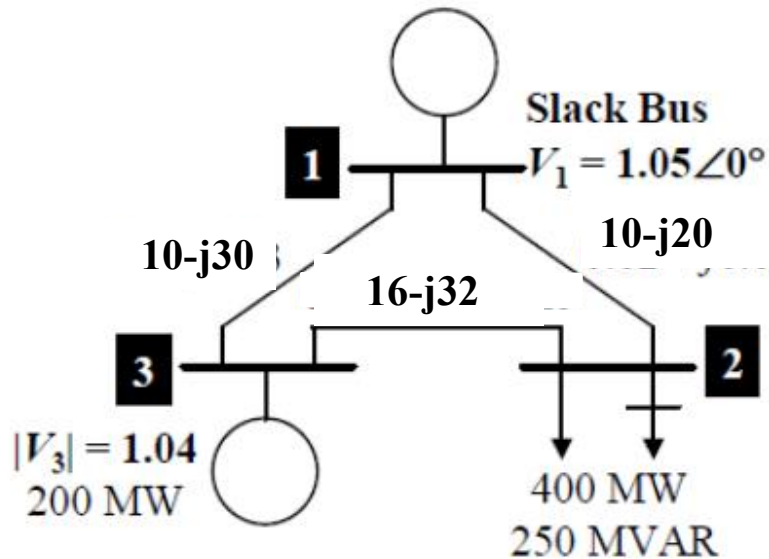
After several iterations, the bus voltages converge to $V_2 = 0.97168 \angle -2.6948^\circ \text{ pu}$ and $V_3 = 1.04 \angle -0.498^\circ \text{ pu}$ and the apparent power for Bus 1 and 3 are found to be $S_1 = 2.1842 + j1.4085 \text{ pu}$ and $S_3 = 2.0 + j1.4617 \text{ pu}$ respectively.

Obtain the line flows, line losses, and the power flow diagram.



Gauss-Seidel Power Flow Solution

Solution:



Line impedances are converted to admittances

$$y_{12} = 10 - j20, y_{13} = 10 - j30 \quad y_{23} = 16 - j32$$

$$V_1 = 1.05 \angle 0^\circ \text{ pu}$$

$$V_2 = 0.97168 \angle -2.6948^\circ \text{ pu}$$

$$V_3 = 1.04 \angle -0.498^\circ \text{ pu}$$

$$\begin{aligned} I_{12} &= y_{12}(V_1 - V_2) = (10 - j20)[(1.05 + j0) - (0.971 - j0.046)] \\ &= (10 - j20)(0.079 + j0.046) \\ &= 1.71 - j1.12 \text{ pu} \end{aligned}$$

$$I_{21} = -I_{12} = -1.71 + j1.12 \text{ pu}$$

$$I_{13} = y_{13}(V_1 - V_3) =$$

$$I_{31} = -I_{13} =$$

$$I_{23} = y_{23}(V_2 - V_3) =$$

$$I_{32} = -I_{23} =$$

The line flows are

$$\begin{aligned} S_{12} &= V_1 I_{12}^* = (1.05 + j0)(1.71 + j1.12) \\ &= 1.7955 + j1.176 \text{ pu} \\ &= 179.55 \text{ MW} + j117.6 \text{ MVAR} \end{aligned}$$

$$S_{21} = V_2 I_{21}^* =$$

$$S_{13} = V_1 I_{13}^* =$$

$$S_{31} = V_3 I_{31}^* =$$

$$S_{23} = V_2 I_{23}^* =$$

$$S_{32} = V_3 I_{32}^* =$$

Gauss-Seidel Power Flow Solution

Solution:

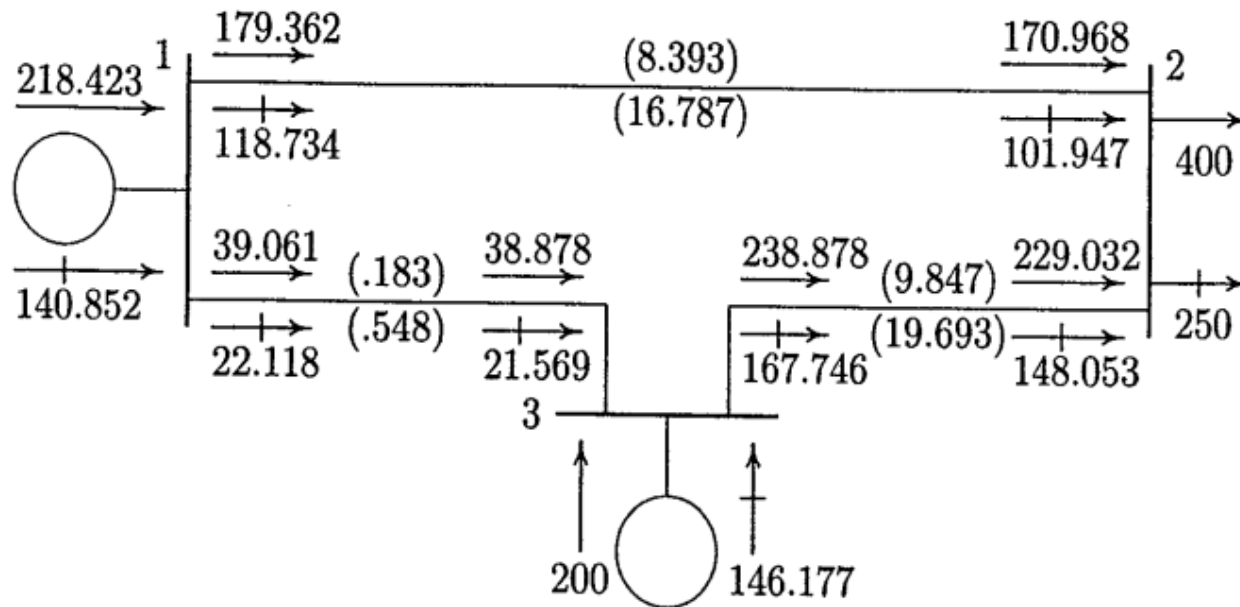
Line flows and line losses are computed as below

$$S_{12} = 179.36 + j118.734 \quad S_{21} = -170.97 - j101.947 \quad S_{L12} = 8.39 + j16.79$$

$$S_{13} = 39.06 + j22.118 \quad S_{31} = -38.88 - j21.569 \quad S_{L13} = 0.18 + j0.548$$

$$S_{23} = -229.03 - j148.05 \quad S_{32} = 238.88 + j167.746 \quad S_{L23} = 9.85 + j19.69$$

The power flow diagram is as below



Reference:

Power System Analysis, Hadi Saadat, McGraw-Hill, Chapter 6.



QUESTIONS AND DISCUSSIONS

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